



Resolving Compositional Features of Solar Wind Source Regions near the Heliospheric Current Sheet at 0.3 au

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Abstract

The Heavy Ion Sensor in Solar Wind Analyser suite on Solar Orbiter has provided unprecedented high-resolution elemental and ionic signatures around the heliospheric current sheet (HCS) from a vantage point near the Sun. Its high cadence time-series observations are used to determine the magnetic connectivity and map these complex transitions, revealing embedded processes of the observed signatures around the HCS crossing at ~ 0.3 au. Electron strahl dropouts observed before the HCS crossing are accompanied by distinct local drops in the oxygen–proton temperature ratio (T_O/T_p), while carbon and oxygen charge-state ratios remain nearly invariant, with at most slight reductions depending on the interval. In contrast, near the true sector boundary (TSB) the heavy-ion composition profiles remain nearly constant, despite marked changes in magnetic and strahl topology. These measurements show that small-scale HCS-related structures at 0.3 au can be systematically matched with their heavy-ion thermal and compositional signatures. A gradual decrease of the Fe/O ratio toward the TSB is observed, reaching a minimum of $(4.54 \pm 0.54) \times 10^{-2}$ at the sector boundary and recovering within a few hours, establishing a clear association between Fe/O depletion and the HCS. After the HCS crossing, T_O/T_p frequently exceeds the super-mass-proportional threshold and is positively correlated with O^{7+}/O^{6+} , while C^{6+}/C^{5+} shows a negative correlation, indicating distinct properties of fully striped carbon within the same sector. As these signatures undergo minimal distortion during radial propagation and evolution, the observations presented in this study offer critical insights into the plasma provenance and the underlying physical processes governing the HCS environment.

Unified Astronomy Thesaurus concepts: Heliosphere (711); Solar wind (1534); Solar corona (1483)

1. Introduction

The heliospheric current sheet (HCS) is the largest coherent structure in the heliosphere, representing the border across which the Sun’s magnetic field reverses polarity from one hemisphere to the other. This boundary originates from the tips of helmet streamers, carrying plasma that is highly fractionated and ionized, making it a critical interface for studying magnetic reconnection and the origins of the slow solar wind (e.g., E. J. Smith 2001; N. M. Viall & J. E. Borovsky 2020). The HCS is typically embedded within the slow solar wind, with plasma velocities between 250 and 500 km s^{−1}. Because the HCS is a region of depressed magnetic field (approaching zero at the neutral sheet), it is characterized by a significant spike in β (D. Winterhalter et al. 1994). Unlike the high-speed wind originating in coronal holes, the slow wind is closely linked to helmet streamers, large bright coronal structures typically observed during eclipses (e.g., M. L. Loucif &

S. Koutchmy 1989; V. Rušin et al. 2013) or through coronagraph imaging (e.g., M. J. Koomen et al. 1998; B. Decraemer et al. 2019). The HCS is also often associated with periodic plasma blobs (e.g., N. R. Sheeley et al. 1997; Y.-M. Wang et al. 1998; E. K. J. Kilpua et al. 2009), which confirm its origin in the hot, confined streamer belt plasma by connecting remote-sensing and in situ observations (A. P. Rouillard et al. 2010a, 2010b).

Near-Sun HCS observations from Parker Solar Probe (PSP; N. J. Fox et al. 2016) and Solar Orbiter (D. Müller et al. 2020) reveal that the structure is more complex than the simple, thin magnetic boundary observed at 1 au. For example, it is frequently embedded within a dense, slow-moving heliospheric plasma sheet (HPS) characterized by high- β plasma and significant variability (B. Lavraud et al. 2020; A. Szabo et al. 2020). The β profiles are highly structured and often show multiple subpeaks or bifurcations, indicating the HCS comprises many small-scale current sheets (S. Eriksson et al. 2022). Therefore, the HCS observations often consist of multiple partial crossings or subsheets rather than a single discrete reversal in magnetic field direction, likely due to the presence of magnetic islands or local ripping of the current sheet (S. D. Bale et al. 2019). The discoveries of the



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reconnection exhaust alongside small-scale magnetic flux ropes in the HCS at 1 au (J. T. Gosling et al. 2005) and near the Sun (T. D. Phan et al. 2022) support the idea that the HCS is prone to tearing-mode instabilities (G. S. Lakhina & K. Schindler 1983) as it expands from the streamer tips into the interplanetary medium (P. C. Liewer et al. 2003; E. Sanchez-Diaz et al. 2019; B. Lavraud et al. 2020; N. Poirier et al. 2020). Properties such as ion densities (N. M. Viall et al. 2009) and heavy-ion charge states (L. Kepko et al. 2016) also exhibit periodic oscillations with timescales ranging from 0.5 to 3 hr, reinforcing their origin in intermittent reconnection or plasma ejection processes near helmet streamers.

The exact production mechanisms responsible for generating these networks of multiple current sheets remain a subject of ongoing debate. While these structures are frequently attributed to wave-like signatures or multiple helmet streamers extending from the corona (N. U. Crooker et al. 1993), potentially as byproducts of sequential magnetic reconnection at the streamer tips (E. Sanchez-Diaz et al. 2019), conclusive physical evidence remains elusive. This variability highlights the HCS’s connection to highly variable sources on the Sun, including small-scale transient events that deposit energetic and fractionated plasma into the heliospheric system. Small-scale compositional observations near the Sun provide a clue to distinguish between these models because the ion composition acts as a unique fingerprint tracing back to the distinct heating, freeze-in, and fractionation processes at the solar origin.

The heavy-ion composition of the HCS is another feature that clearly marks its coronal streamer belt origin. Plasma within these streamers is hotter, denser, and more fractionated than in coronal holes (J. C. Raymond et al. 1997a), resulting in distinct compositional (L. Zhao et al. 2009) signatures within the HCS. Slow solar wind plasma exhibits elevated levels of elements with low first ionization potential (FIP), such as iron, compared to high-FIP elements like oxygen. This fractionation likely occurs within the chromosphere, where low-FIP ions are preferentially transported upward by plasma flows along magnetic field lines (J. M. Laming 2015). As such, enhanced Fe/O indicates slow wind (F. Ipavich et al. 1992), where plasma originates from open magnetic fields that were previously closed (J. M. Laming et al. 2019). Enhanced higher ionic charge states, such as an increase in the O^{7+}/O^{6+} ratio, signify that the plasma originates in high-temperature and high-density streamer belt regions (M. R. Aellig et al. 1999; T. H. Zurbuchen et al. 2002; L. Zhao et al. 2009). Conversely, the alpha particle abundance relative to protons is lower in the HCS, contrasting with the alpha-rich fast wind from coronal holes (N. U. Crooker et al. 1996; B. L. Alterman & R. D’Amicis 2025a, 2025b; B. L. Alterman et al. 2025). These compositional markers serve as critical diagnostics from in situ plasma observations around the HCS for identifying both the source regions and physical conditions through which plasma flows into the interplanetary medium because they are thought to freeze-in near the Sun and remain largely invariant through the heliosphere (T. H. Zurbuchen et al. 2002; E. Landi et al. 2014).

In situ measurements have revealed that small-scale heavy-ion signatures are a common feature near the HCS at 1 au. For instance, the preferential depletion of fully stripped ions (C^{6+}) relative to C^{5+} has been observed as the so-called outlier solar wind, occurring both within the slow solar wind

(L. Zhao et al. 2017a) and inside coronal mass ejections (M. Kocher et al. 2017), while showing spatial correlations with the current sheet (Y. J. Rivera et al. 2021). Additionally, short-term, intermittent mass-fractionated heavy-ion dropouts have been detected near sector boundaries at 1 au, attributed to gravitational settling within large, closed coronal loops where plasma remains trapped long enough for heavy ions to sink before being released (M. J. Weberg et al. 2012). However, identifying the origins of these features and establishing their definitive connection to the HCS is highly ambiguous at 1 au because fine-scale structures undergo significant deformation during propagation. Further, several heavy-ion signatures, especially for thermal profiles, remain susceptible to modification by Coulomb collisions with protons (P. J. Tracy et al. 2015). Compositional observations near the Sun bypass these transport effects, providing the confidence to unambiguously link these structures to their source regions and give restrictions on processes and mechanisms at work.

Resolving small-scale features, which trace back to the dynamic source regions, requires high time resolution observations in the inner heliosphere, which are now only available through missions such as PSP and Solar Orbiter. The instrumentation aboard Solar Orbiter, particularly the Heavy Ion Sensor in the Solar Wind Analyser suite (SWA-HIS, C. J. Owen et al. 2020; S. A. Livi et al. 2023; S. A. Spitzer et al. 2026, in preparation), enables the measurement of HCS-related plasma composition features as close as 0.3 au, including heavy-ion charge states and fractionated coronal plasma. This study aims to explore the interconnected phenomena governing the evolution and dynamics of the HCS by exploring its compositional and structural properties during a dynamic HCS crossing event at 0.3 au.

2. Methodology

2.1. Solar Orbiter Data

All three sensors of the SWA suite (C. J. Owen et al. 2020) on Solar Orbiter are used to analyze the HCS crossing event in coordination with the Solar Orbiter Magnetometer (MAG; T. S. Horbury et al. 2020) for the interplanetary magnetic field vectors. The solar wind proton parameters (velocity, number density, and kinetic temperature) are derived from measurements by the Proton-Alpha Sensor (SWA-PAS). The strahl electron pitch angle distributions (PADs) are derived from measurements by the Electron Analyser System (SWA-EAS). All of the data used in the analysis are available as level 2 or 3 data products from the Solar Orbiter Archive (SOAR, <https://soar.esac.esa.int/>). Solar wind parameters and magnetic field vectors are used as downloaded from the open data archives (SOAR, also available from Space Physics Data Facility, SPDF, managed by National Aeronautics and Space Administration, NASA). When processing the SWA-EAS level 3 PADs, the phase-space density (PSD) in the 120–180 eV energy range are summed over four energy bins to approximate the strahl component of the solar wind electrons. The iron-to-oxygen ratio (Fe/O), oxygen charge-state ratio (O^{7+}/O^{6+}), and carbon charge-state ratio (C^{6+}/C^{5+}) and O^{6+} temperature (T_O) are from SOAR’s level 3 data products derived from measurements by SWA-HIS.

2.2. Alfvénicity and Normalized Cross Helicity

Two physical parameters are applied in this study by using magnetic field and proton velocity vectors to identify the nature of the HCS crossing: the Alfvénicity and normalized cross helicity (σ_C). The Alfvénicity describes the degree of correlation between the velocity field fluctuations ($\delta\mathbf{v}$) and the magnetic field fluctuations ($\delta\mathbf{b}$). Since the relationship for ideal Alfvénic fluctuations is $\delta\mathbf{v} = \mp\delta\mathbf{b}$ (in Alfvén units), the correlation coefficients between each of the vector components show the degree of the dominance of Alfvénic turbulence in the observed fluctuations. For each observation window, the vector components are defined as follows:

$$\delta\mathbf{b} = \mathbf{B} - \langle\mathbf{B}\rangle, \quad (1)$$

$$\delta\mathbf{V} = \mathbf{V} - \langle\mathbf{V}\rangle, \quad (2)$$

and cross-correlation coefficients are calculated using the three vector components of the velocity and magnetic field (r, t, n). The solar wind is dominated by large-amplitude Alfvén waves propagating outward from the Sun, and strong correlations are typically found in the fast wind at 1 au (e.g., J. W. Belcher & L. Davis 1971). The slow solar wind can be Alfvénic especially during the solar maximum (R. D’Amicis et al. 2011). The Alfvénic slow winds are thought to have a similar origin to the fast wind, for example, from the isolated coronal holes in the equatorial region (R. D’Amicis & R. Bruno 2015).

The σ_C of the solar wind perturbations is a dimensionless parameter that is used to measure the alignment and relative dominance of Alfvénic fluctuations in solar wind plasma, ranging from -1 to $+1$ (e.g., D. A. Roberts et al. 1987). The fast wind $|\sigma_C|$ tends toward ~ 1 , while the $|\sigma_C|$ trend decays quickly for slow wind (B. L. Alterman & R. D’Amicis 2025a). In general, σ_C decreases with radial distance as the wind expands (D. A. Roberts et al. 1987), and this trend is most evident in low $|\sigma_C|$ wind (E. Marsch & C.-Y. Tu 1990). This evolution is driven by inward-propagating waves generated through velocity shear at stream interfaces and reflection of large-scale gradients, leading to a more balanced turbulent state where $|\sigma_C|$ approaches zero (M. Dobrowolny et al. 1980; B. Bavassano et al. 1998). Modern high-resolution data from the inner heliosphere have allowed further refinement of this scenario, revealing that while σ_C remains high in the inner corona, it is punctuated by switchbacks that maintain the Alfvénic correlation (M. D. McManus et al. 2020; D. Perrone et al. 2025). The σ_C is calculated using the same interplanetary magnetic field and solar wind velocity datasets used for the Alfvénicity, resulting in the following formula:

$$\sigma_C = \frac{2 \langle \delta\mathbf{V} \cdot \delta\mathbf{B} \rangle / \sqrt{\mu_0 \langle \rho \rangle}}{\langle \delta V^2 + \delta B^2 / (\mu_0 \langle \rho \rangle) \rangle}, \quad (3)$$

where μ_0 and ρ denote vacuum permeability and plasma mass density, respectively. When deducing the Alfvénicity and σ_C , the magnetic field data (1/8 s cadence) are averaged over PAS’s solar wind velocity data (4 s cadence) to calculate the components. Then, measurement windows of level 3 SWA-HIS data (~ 10 minutes) are applied for averaging each value that matches to the SWA-HIS observation cadences.

2.3. Source Region Signatures

To map in situ observations to their source region, we use Air Force Data Assimilative Photospheric Flux Transport (ADAPT) magnetograms (C. N. Arge et al. 2010) to determine the surface magnetic field distribution on the Sun’s surface. The coronal hole regions and magnetic field polarities are identified using MHDweb (Predictive Science Inc., <http://www.predsci.com/mhdweb/home.php>) in the context of the magnetohydrodynamic around a sphere (MAS) model. The footprints and connectivity of Solar Orbiter are calculated using the Magnetic Connectivity Tool (<http://connect-tool.irap.omp.eu>) (A. P. Rouillard et al. 2020). The coronal magnetic field is reconstructed by the tool using the potential field source surface (PFSS) model and then extended into the heliosphere assuming a Parker spiral model. The solar wind can then be mapped ballistically back to a source surface at the height of $\sim 3 R_\odot$ (P. Riley & R. Lionello 2011). To correctly back-map the events observed in situ to those by remote-sensing instruments, the delay between the two observations of the same event are taken into account. The Magnetic Connectivity Tool also provides approximations to the HCS boundaries accounting for the solar wind speed at Solar Orbiter.

3. Results

3.1. Solar Orbiter’s HCS crossing on April 13

Figure 1(a) shows the position of the spacecraft on 2023 April 13 (day of year (DOY) 103). The HCS crossing of Solar Orbiter occurred at 7:30 UT 2023 April 13 during the outbound leg of its orbit. Over the period April 12–13, the spacecraft covered heliocentric distances of 0.297–0.303 au and latitudes between -3.88° and -3.61° . Figure 1(b) tracks the position of Solar Orbiter mapped on the Sun’s surface magnetic field properties in the Carrington rotating frame ± 6 hr before and after the HCS crossing at 6:00 UT on 2023 April 13. The surface magnetic field observations are shown as a map for the Carrington rotation 2269. Coronal hole regions with magnetic field polarities (blue: toward the Sun, red: away from the Sun) overlaid on the magnetogram map identify the dominant polarity of the radial field in the sector. The thick black line separates two regions with opposite radial fields, indicating the expected HCS boundary calculated by the Magnetic Connectivity Tool. Subspacecraft points (cross marks) connect to the magnetic field footprints (circles) in the same color. These subspacecraft points stay essentially connected to the boundary of the HCS. The in situ radial magnetic field observed by MAG supports this observation because the magnetic field is predominantly negative throughout the time period. The magnetic footprints also stay in a similar area, while they split before (blue) and after (red) the period of interest. This feature indicates the connectivity point moved from the negative B_r region to the positive B_r region, as supported by the reversed radial field geometry. As expected, the computed footprint location at the moment of the HCS crossing did not converge into one solution because of the large loop structure of the magnetic field connected to both polarities of the radial field near the HCS.

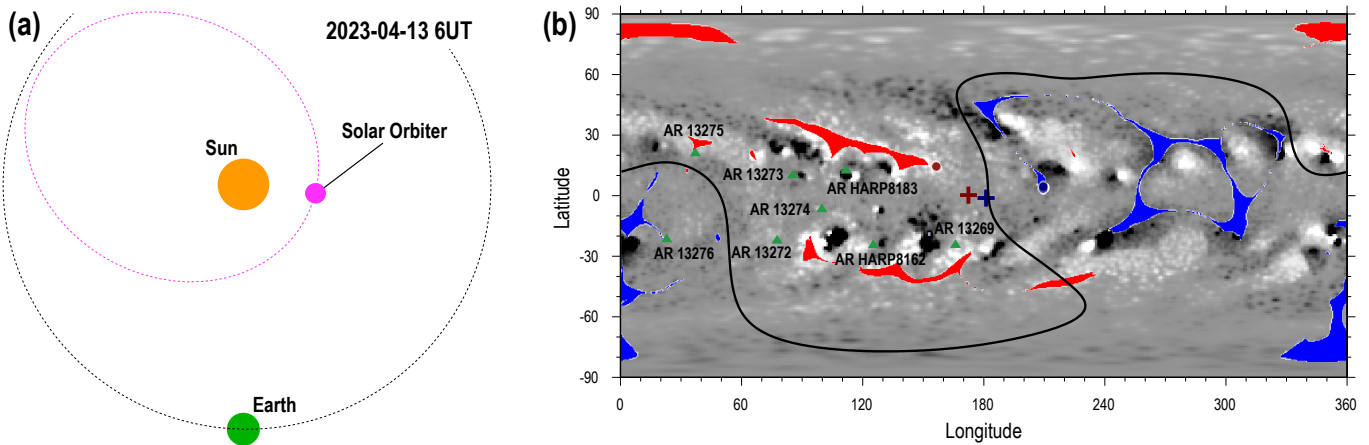


Figure 1. (a) Schematic diagram showing locations of the Earth and Solar Orbiter in the solar ecliptic coordinate system during the HCS crossing focused on in this paper. (b) ADAPT magnetogram for the Carrington rotation 2269 with spacecraft footprints and estimated HCS. The coronal holes are also marked with red (blue) for positive (negative) B_r . The spacecraft locations before (blue) and after (red) 6 hr of HCS crossings are marked by the crosses with the magnetic footprints with circles. The expected HCS location is also indicated by the black line. The magnetic footprints are calculated by the magnetic connectivity tool. The green triangles denote locations of active regions identified in CR2269.

3.2. Structures of the HCS

Figure 2 summarizes the in situ context of Solar Orbiter’s HCS crossing during DOY 102-103, 2023, including the plasma properties, the magnetic field and solar wind signatures, and compositional signatures derived from heavy ions. A joint analysis of the PAD, magnetic field, density, and velocity field reveals signatures of the HCS crossing.

The first panel (Figure 2(a)) shows the plasma beta (β), defined as the ratio of plasma thermal pressure (represented here by proton pressure due to the dominance) to magnetic pressure ($P_{\text{plasma}}/P_{\text{mag}}$). The Solar Orbiter observations show highly structured plasma β signatures, as found in previous observations in the inner heliosphere. The HCS crossing presented here is not a simple current sheet but a succession of alternating high- β and low- β regions with structures in multiple layers. The solar wind and magnetic field parameters are summarized in Figures 2(b)–(e). The radial magnetic field (B_r) reached zero multiple times in shorter timescales, while the general trend shows that B_r transitioned from negative (Region 1, blue) to positive (Region 4, gray). This global trend of the B_r reversal, an important feature of HCS crossings, is consistent with the connectivity analysis in Figure 1(b).

The electron strahl PAD is shown in Figure 2(f). Broadly, the change in the PAD of electron strahl from 180° to 0° was observed from 7 to 8 UT on DOY 103, which indicates the true sector boundary (TSB) during the HCS crossings. The region near the timing of TSB crossing is identified as HCS in Figure 2 and marked as Region 3 (red area). Taking a closer look, this region is separated into two regions by the B_r reversal indicated by the vertical red dashed line, which drastically changes the magnetic field connectivity to the Sun. In the PAD, the counterstreaming electrons were found in the first half of Region 3, while a strahl electron dropout (SDO) was observed in the second half. Before the TSB, the PAD shows three instances of SDO (Region 2, green). Such SDOs indicate that X-lines are in the upstream of Solar Orbiter closer to the Sun and Solar Orbiter is located in a local magnetic blob (e.g., M. J. Owens & R. J. Forsyth 2013). Although the density increase was fully observed in the third SDO, the first and second were accompanied by only partial (or no) increase in

density. The second SDO remained consistently within a low β regime. Considering the constant total magnetic field in the SDO, the trend suggests detachment from nearby high-density regions.

Figure 2(g) plots the Alfvénicity. Typically, high Alfvénicity indicates that the local solar wind originates in open field lines in coronal holes at 1 au (S. L. Yardley et al. 2024). In contrast, the data for this entire event shows relatively high Alfvénicity overall at 0.3 au, which is unexpected given that the HCS is connected to equatorial structures with intermittently closed magnetic fields. Positive Alfvénicity indicates waves propagating outward from the Sun under the radial field configuration with $B_r < 0$. After the HCS crossing, the radial field configuration switched to $B_r > 0$ while the wave is still expected to propagate outward from the Sun. Therefore, an Alfvénicity reversal also occurs after the observed HCS crossing in Region 4. Within the HCS (Region 3), the Alfvénicity remained positive before and after the B_r reversal. This behavior indicates the existence of sunward-propagating wave in the second half of Region 3 within the magnetic isolated blob.

Figure 2(h) plots the normalized cross helicity (σ_C). Similar to the Alfvénicity, $|\sigma_C|$ being close to 1 indicates that the solar wind originates on radial magnetic field lines in coronal holes (B. L. Alterman & R. D’Amicis 2025a, 2025b). Overall, the σ_C changed polarity after the HCS crossing from negative to positive. Within Region 3, the reversal of σ_C matched the timing of B_r reversal, in contrast to the Alfvénicity case. Therefore, the second half of Region 3 could be characterized as an area dominated by sunward-propagating Alfvén waves. However, the polarity of σ_C reversed in the second SDO, showing intermittent connectivity to the opposite side of the sector boundary ($B_r > 0$ side) within this period. Considering the low- β , as described above, and the relatively low strahl electron flux, this region might be freshly connected to the other sector through magnetic reconnection near the current sheet. The third SDO included mild and intermittent strahl electron signatures, and nearly counterstreaming profiles. There was a short period of σ_C reversal accompanied by the electron PAD reversal as well, but the reversal signature was

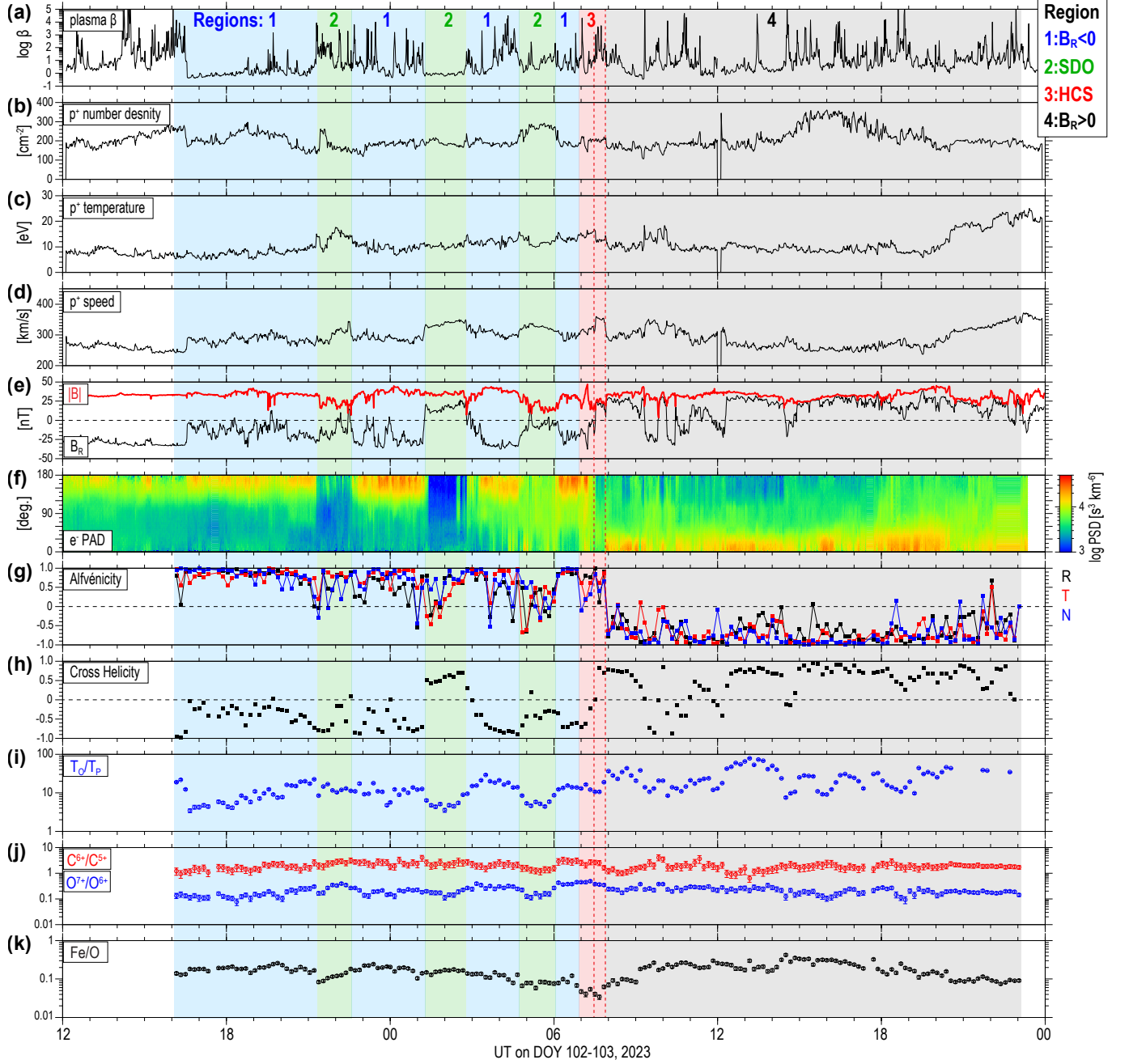


Figure 2. A summary of in situ measurements during the HCS crossing of Solar Orbiter from 2023 April 11 to 2022 April 13 by the SWA and MAG instruments. (a) Plasma β based on SWA and MAG. (b)–(d) Proton number density N_p , temperature T_p , and velocity V_p as measured by SWA-PAS. (e) Total ($|B|$, red) and radial (B_r , black) components of the interplanetary magnetic field (nT) measured by MAG. The dashed line signifies 0 nT. (f) PAD of electrons measured by SWA-EAS. The color scale shows the electron PSD. (g) Alfvénicity evaluated by the correlation coefficient of δB and δV in Radial (R), Tangential (T), and Normal (N) coordinates. The dashed line indicates a correlation coefficient of 0. (h) Cross helicity between B and V . (i) O^{6+} temperature normalized by the proton temperature reported by PAS (blue). (j) Charge-state ratios for C^{6+}/C^{5+} (red) and for O^{7+}/O^{6+} (blue). (k) Fe/O ratio. Three strahl dropouts (SDO) regions are shaded in green while the rest of the $B_r < 0$ region of interest is shaded in blue. The region near the true sector boundary is shaded in red and marked as HCS. The two red dashed lines represent the reversals of cross helicity and Alfvénicity, respectively. The $B_r > 0$ region is shaded in gray.

not as evident as the second SDO. On the other hand, no clear evidence of full reversal of σ_C was found in the first SDO. This suggests that the magnetic field lines are reconnected within the same sector at the location far from the TSB. All SDOs were more or less accompanied by the signatures of increasing proton speeds, consistent with reconnection jet. Note that short-term σ_C reversals were also found a few times after the HCS crossing in the $B_r > 0$ regions (9:00–11:00 UT).

3.3. Ionic Composition in the HCS

Figures 2(i)–(k) summarize the temporal variations of heavy-ion properties. In this study, we focus on four parameters: oxygen to hydrogen temperature ratio (T_O/T_p , Figure 2(i)), O^{7+}/O^{6+} ratio (Figure 2(j), blue), C^{6+}/C^{5+} ratio (Figure 2(j), red), and Fe/O ratio (Figure 2(k)). The T_O/T_p ratios are calculated based on the isotropic Oxygen (O^{6+}) temperature, derived from SWA-HIS data, normalized by the

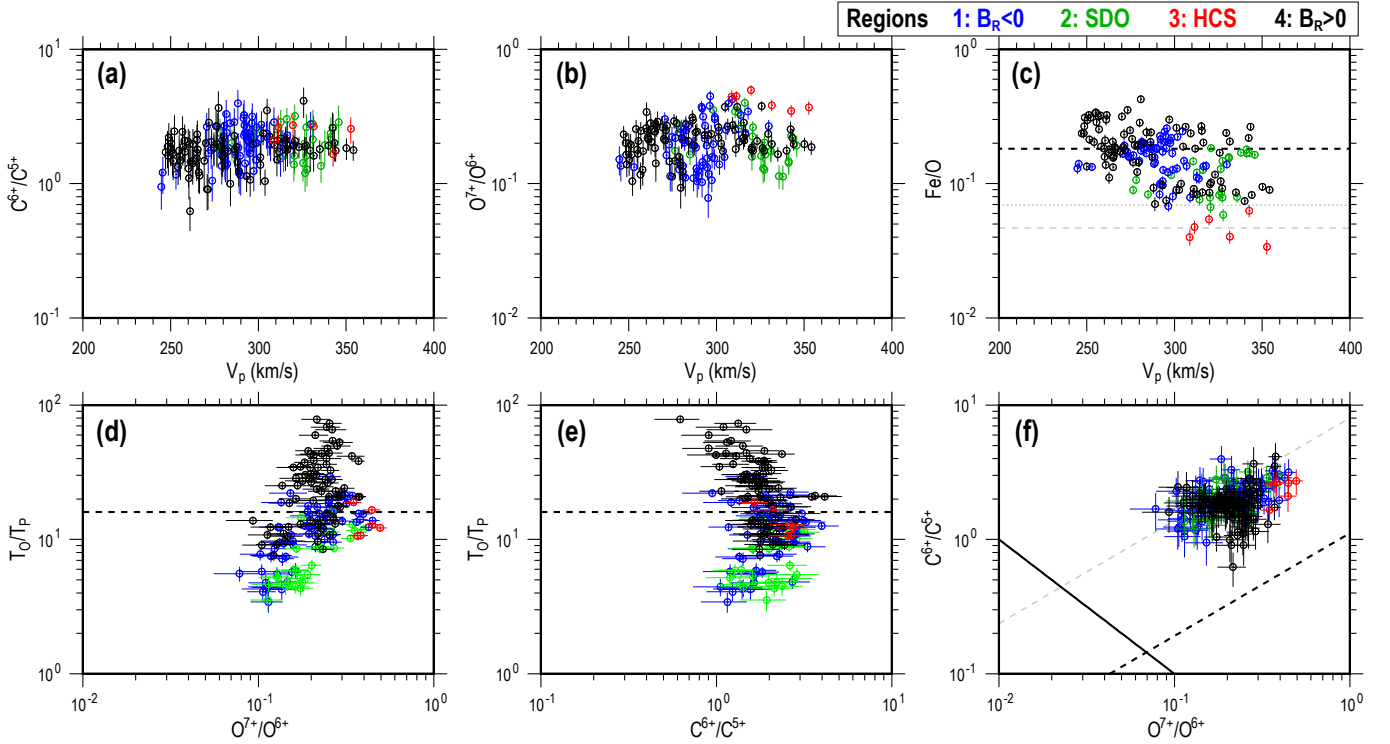


Figure 3. Scatter plots to compare key heavy-ion parameters during the entire HCS crossing event at 0.3 au. The colors highlight data from each region defined in Figure 2 (blue: $B_r < 0$, green: SDO, red: HCS, and black: $B_r > 0$). (a), (b) Comparison of proton velocity and C^{6+}/C^{5+} and O^{7+}/O^{6+} ratios, respectively. (c) Comparison of proton velocity and Fe/O ratio. Typical values calculated from SOHO/UVCS observations are indicated by horizontal lines (black dashed line: coronal value by U. Feldman & J. M. Laming 2000; gray dotted line: photosphere values by M. Asplund et al. 2009; gray dashed line: photosphere value by N. Grevesse & A. J. Sauval 1998). (d), (e) Comparison of proton to O^{6+} temperature ratio as a function of O^{7+}/O^{6+} and C^{6+}/C^{5+} ratios, respectively. The super-mass-proportional limit ($\times 16$) is indicated by the horizontal dashed line. (f) Comparison of C^{6+}/C^{5+} and O^{7+}/O^{6+} ratios. The average value and outlier wind limit from L. Zhao et al. (2017a) are indicated by black and gray dashed lines. The black line on the bottom left corner shows the streamer and coronal hole boundary determined by 1 au observation (R. von Steiger & T. H. Zurbuchen 2015).

proton temperature, derived from SWA-PAS data, during the relevant time window.

Figure 2(i) plots the T_O/T_p profile throughout the HCS crossing event. Among the parameters evaluated, T_O/T_p exhibited the most significant fluctuations throughout the duration of the crossing. T_O/T_p dropped in the second and third SDOs, while the C^{6+}/C^{5+} and O^{7+}/O^{6+} ratios (Figure 2(j)) showed mostly small variations before and after the HCS crossing. The T_O/T_p ratio was relatively higher after crossing the HCS (Region 4). In most such cases, T_O/T_p reached beyond super-mass-proportional (>16). In Regions 1 and 2, both C^{6+}/C^{5+} and O^{7+}/O^{6+} gradually increased toward the TSB and decreased within the third SDO. After passing TSB, in Region 4, the C^{6+}/C^{5+} increased at the timing when T_O/T_p slightly dipped around 10:00 UT, while O^{7+}/O^{6+} stayed nearly constant until 14:00 UT. The C^{6+}/C^{5+} and O^{7+}/O^{6+} showed a trend of anticorrelation afterwards. This sector contains total seven active regions (Figure 1(b)); the elevated T_O/T_p ratio likely originates from these features.

The transition from fast to slow wind during HCS crossings is often marked by an increase in Fe/O due to the FIP bias at 1 au. The decreasing trend after 15:00 UT on DOY 103 seems to reflect such a FIP bias effect in response to the increasing solar wind speed. However, the observations near the TSB revealed an inverse trend on the Fe/O profile compared to the expectation from the typical slow wind. As illustrated in Figure 2(k), Fe/O gradually decreased toward the HCS, reaching a local minimum within the HCS (Region 3). Fe/O slightly varied across the boundaries characterized by the

Alfvénicity and σ_C reversals (vertical dashed lines in Figure 2), while the values remained the lowest level in the entire crossing. The Fe/O value stayed around 0.16 in the regions around the HCS (Regions 1 and 4), which is slightly higher than the typical slow solar wind value of 0.12 (F. Ipavich et al. 1992). The gradual increases in Fe/O are observed before and after the core of the dip, which caused these higher Fe/O ratios. Such lower Fe/O recovered to the original level in a few hours at the beginning of Region 4. The trend of Fe/O was not significantly affected by the solar wind sectors or speed jumps other than a few exceptions (e.g., 21:20 UT on DOY 102). During the Fe/O dip, the solar wind was slow, varying within the range 250–350 km s⁻¹, concurrent with the Fe/O reduction.

Figure 3 summarizes the correlations among key parameters in scatter plots. The three panels in the first row (Figures 3(a), (b), and (c)) plot the correlation of C^{6+}/C^{5+} , O^{7+}/O^{6+} , and Fe/O as a function of (proton) solar wind speed. During the HCS crossing, both carbon and oxygen charge-state ratios show small variations within the observation time frame and remain within the typical slow-wind regime. These results are reasonable considering that the solar wind speed varies only within 250–350 km s⁻¹. Note that both carbon and oxygen ratios are somewhat higher than the typical values in the slow-wind range, consistent with the solar maximum cases for the slow wind (D. Koutroumpa 2024). In the SDOs (Region 2) and the HCS (Region 3), the solar wind speed is actually higher than the other regions by ~ 100 km s⁻¹. The increase in

Table 1
SWA-HIS Observations of the Fe/O Ratio at 0.3 au

	Regions	Fe/O Ratio
1	Br < 0	$(1.57 \pm 0.13) \times 10^{-1}$
2	SDO	$(1.12 \pm 0.10) \times 10^{-1}$
3	HCS	$(4.54 \pm 0.54) \times 10^{-2}$
4	Br > 0	$(1.56 \pm 0.15) \times 10^{-1}$

Note. The data are averaged over each region.

O^{7+}/O^{6+} in the HCS with respect to the SDOs differentiates the plasma properties between these regions.

Figure 3(c) is a scatter plot of Fe/O as a function of the solar wind speed. The black dashed line shows the coronal reference value (U. Feldman & J. M. Laming 2000), and the two gray horizontal lines show the photospheric reference values obtained from remote-sensing observations by the Ultraviolet Coronagraph Spectrometer (UVCS; J. L. Kohl et al. 1995) on the Solar and Heliospheric Observatory: 4.7×10^{-2} N. Grevesse & A. J. Sauval (1998) and 6.5×10^{-2} M. Asplund et al. (2009). Over the observed time frame, Fe/O exhibits a weak dependence on solar wind speed. In the HCS, where the speed lies between 300 and 350 km s⁻¹, Fe/O reaches values comparable to, or lower than, the photospheric references. The Fe/O values within the SDO intervals are slightly reduced relative to adjacent regions, while Regions 1 and 4 essentially show the same Fe/O level. Table 1 summarizes the averaged values of Fe/O observations from this study for all four regions. The averaged Fe/O in Region 3 was $\sim 4.5 \times 10^{-2}$, which equates to these photospheric values.

The second row in Figure 3 shows the oxygen preferential-heating properties. The first two figures in the second row (Figures 3(d), (e)) show the correlation of O^{7+}/O^{6+} and C^{6+}/C^{5+} with the oxygen (O^{6+}) to proton temperature ratio (T_O/T_p). For Regions 1–3, the O^{6+} temperature remained below mass-proportional temperature (horizontal dashed lines). However, the trend was notably different between the carbon and oxygen cases in Region 4. For oxygen (Figure 3(d)), T_O/T_p has a positive correlation with O^{7+}/O^{6+} beyond super-mass-proportional temperature. This implies either that a common mechanism drives both O^{6+} preferential heating (over proton) and the ionization of oxygen (e.g., high coronal temperature) below the freeze-in height or that the source of O^{6+} preferential heating could be collocated with the higher coronal temperature region.

Conversely, the T_O/T_p ratio exhibits a negative correlation with C^{6+}/C^{5+} in Region 4 (Figure 3(e)), where the oxygen temperature exceeds the mass-proportional threshold for most of the interval. In contrast, across the other regions (Regions 1–3) with T_O/T_p ratio below the mass-proportional threshold, the C^{6+}/C^{5+} ratio correlates positively with T_O/T_p . This result indicates the loss of fully stripped carbon (C^{6+}) over C^{5+} , and the trend of C^{6+}/C^{5+} bifurcates from that of O^{7+}/O^{6+} beyond the oxygen mass-proportional heating threshold.

Figure 3(f) shows the O^{7+}/O^{6+} to C^{6+}/C^{5+} ratio in all regions. The diagonal black line in the bottom left corner of the figure shows the boundary between the streamer and the coronal hole wind (R. von Steiger & T. H. Zurbuchen 2015). The observed values remained on the higher side of the

streamer-like solar wind for the entire observation period. The gray dashed line shows the averaged trend, while the black dashed line indicates the boundary for the outlier slow wind, which indicates the appearance of considerable amount of fully stripped carbons, both inferred from L. Zhao et al. (2017b). As illustrated in Figure 3(f), the measured C^{6+}/C^{5+} ratio is consistent with the nominal range expected for slow solar wind of streamer origin. The observed values are slightly below the average line, indicating higher O^{7+}/O^{6+} over C^{6+}/C^{5+} overall. For this study, the relative amount of fully stripped C^{6+} depletion is smaller than the reported slow-wind cases, since the measurements fall sufficiently above the dashed line. It is unclear if there are trends to characterize each region in terms of the O^{7+}/O^{6+} and C^{6+}/C^{5+} ratios. Note that the O^{7+}/O^{6+} ratio was among the highest during the HCS crossing (Region 3), showing the enhanced oxygen freeze-in temperature (e.g., J. Geiss et al. 1995; S. Hefti et al. 2000) at the foot of the HCS regions. On the other hand, no clear depletion of C^{6+}/C^{5+} was measured in Region 3 and the oxygen temperature was around the mass-proportional level.

4. Discussion

This section discusses the key findings from SWA-HIS observations during the HCS crossing event at 0.3 au. We focus on two core characteristics of transient small features: the structures within the current sheet and the origin of strahl dropouts, as well as three sector-level features: the Fe/O dropouts near the TSB, the oxygen preferential heating and ionic charge-state ratio, and the fully stripped carbon dropouts. SWA-HIS observations during the HCS crossing at 0.3 au highlight the oxygen–proton temperature ratio T_O/T_p as a key diagnostic of small-scale structure in and around the current sheet. Because oxygen has not yet fully thermally equilibrated with protons at this distance by Coulomb collisions (P. J. Tracy et al. 2015), T_O/T_p still reflects coronal signatures, which is difficult to recover at 1 au.

4.1. The Structures within the Current Sheet

The first and second halves near the TSB exhibited distinct differences in magnetic topology based on the electron strahl signatures. SWA-HIS observations revealed no significant variations in T_O/T_p , C^{6+}/C^{5+} , O^{7+}/O^{6+} , or Fe/O between these two regions. These ionic and elemental signatures are consistent with the interpretation that the sampled plasma has already passed through both the charge-state freeze-in height and the main region of preferential oxygen heating by the time it is structured into these small-scale features. These observations place constraints that the structuring of these features occurs at or beyond the heights where composition and charge states are fixed.

In the first half, the counterstreaming electron strahl indicates that the locations of the reconnection X-line is in the downstream (i.e., antisunward direction) of Solar Orbiter, since both ends of the magnetic field line are still connected to the Sun (J. T. Gosling et al. 1987). The second half’s SDO is consistent with a scenario in which the reconnection X-line lies between the Sun and Solar Orbiter, and Solar Orbiter is located within a magnetically isolated plasma blob. This structure can be interpreted as the nested magnetic loops, still connected to the Sun, overlying the isolated flux rope. These signatures are compatible with smaller structures within the

HCS produced by multiple magnetic reconnection over a range of heliocentric distances. The region within the counter-streaming strahl electrons shows a spike of magnetic field (Figure 2(e)) with lower plasma density (Figure 2(b)), which could be associated with a plasmoid within the HPS. Such small-scale structures near the HCS have been identified at 1 au using multiple spacecraft (C. Foullon et al. 2009, 2011). The SWA-HIS results are consistent with a scenario in which these small-scale structures originate within the HCS, exceeding the height of their source at the helmet streamer tip. We emphasize that, for this single event, the data constrain where such structures are sampled along the flow but do not uniquely determine their coronal formation height.

4.2. The Origin of Strahl Dropouts

The invariant heavy-ion charge-state signatures, coupled with the observed drops in T_O/T_p within the SDOs, suggest that these transient features are sampled downstream of the charge-state freeze-in region but while preferential heating of oxygen is still active or recently ceased. Thus, the observations place constraints on their origin relative to the freeze-in and heating zones, even though the exact mechanism generating the SDOs cannot be uniquely identified.

Previous observations at 1 au suggest that sector boundary encounters are often associated with multiple HCS crossings (e.g., N. U. Crooker et al. 1993; D. Winterhalter et al. 1994; E. J. Smith 2001). The SDOs found in Region 1 could also be categorized as partial HCS crossings because they are accompanied by a decrease in magnetic field and increase in proton speed. The proton speed was increased by 50 km s^{-1} for all SDO regions, which is consistent with the properties in magnetic reconnection outflow within the SDOs, while the velocity increase was milder than the cases closer to the Sun ($<20 R_\odot$) (T. D. Phan et al. 2022). Occasional reductions in Alfvénicity over time periods within Regions 2 and 3 were observed, which is also a typical feature of reconnection outflows, because the magnetic field is always perpendicular to the bulk plasma velocity in the heart of the current sheet (e.g., T. D. Phan et al. 2022).

SWA-HIS observations demonstrated that T_O/T_p dropped in the second and third SDOs, while the C^{6+}/C^{5+} and O^{7+}/O^{6+} ratios showed mostly small variations before the HCS crossing. This feature is consistent with reconnection (or other connectivity-changing processes) occurring above the charge-state freeze-in altitude ($1.0\text{--}2.0 R_\odot$, A. Bürgi & J. Geiss 1986; R. von Steiger 1996; E. Landi et al. 2012) and below the height where O^{6+} is fully heated. We note that the current single-event dataset does not allow us to pin down the exact altitude of these processes, nor to rule out alternative mechanisms. From UV observations, the O^{6+} heating zone is estimated around $3\text{--}5 R_\odot$ (L. Abbo et al. 2010). Following this result, the locations of the reconnection which caused the two SDOs is possibly located below such a heating region, nominally below or near the sonic point. From in situ observations, the Coulomb relaxation model (J. C. Kasper et al. 2017) predicts that the preferential heating is active over an extending zone from the transition region in the lower corona to a boundary in the inner heliosphere (possibly beyond 0.3 au) for minor ion species (J. Holmes et al. 2024). Our results within these two SDOs are consistent with both scenarios.

While the combination of strahl dropout, modest proton speed increases, and the reductions in Alfvénicity are

suggestive of reconnection outflows, the available measurements do not provide a unique diagnostic of reconnection versus other processes. We therefore present reconnection-based scenarios as plausible interpretations constrained by the observed composition and thermal signatures, rather than definitive identifications of the underlying mechanism.

4.3. The Fe/O Dropouts near TSB

The reduction of Fe/O ratio at TSB is not explained by the FIP bias effect alone. A natural explanation is gravitational settling because particles can be confined in closed magnetic field loops for extended periods at the origin of the streamer. The close alignment of the Fe/O minimum with the TSB supports this interpretation.

J. C. Raymond et al. (1997b) and J. C. Raymond et al. (1997a) found that O, Mg, Al, Si, S, and Fe in streamer cores are severely depleted compared to streamer legs, irrespective of the FIP effect. This feature is attributed to a depletion in the oxygen elemental abundance relative to photospheric values (G. Noci et al. 1997; J. C. Raymond et al. 1997a) due to gravitational settling in closed field regions. Compared to quiescent streamers, active-region streamers have a wider range of ion abundances in their cores (M. Uzzo et al. 2006). J. C. Raymond et al. (1997a) attributed this difference to the combination of the FIP effect and gravitational settling, and the higher FIP of oxygen causes the increase of Fe/O in the center (0.158) compared to the value in the leg (0.125). The injection of material into both core and legs with the FIP abundance pattern is expected, while the plasma confinement time of materials between the center and the leg cause different decreases of gravitational settling. Our results show a significantly lower Fe/O ratio in the center of HCS (0.045) with a dip near the HCS, and are thus not fully consistent with reported UV results. The highest values in Regions 1 and 4 are consistent with the center streamer values of UVCS observations, while the gradual depletion of Fe/O toward the HCS was not observed by UVCS.

Conversely, in situ observations at 1 au identified periods where the logarithmic Fe/H ratio dropped at least one standard deviation below its average for at least 4 hr (M. J. Weberg et al. 2012). Considering the O/H ratio in that work, the average Fe/O ratio in the depletion was $(6.55 \pm 0.98) \times 10^{-2}$, which is consistent with our single-event HCS value of $(4.54 \pm 0.54) \times 10^{-2}$. At 1 au, $\sim 50\%$ of these events are associated with the solar wind stream interfaces within 2 days, which suggest the relation with the HCS (typically embedded in the slow-wind side of the stream interaction regions). In SWA-HIS observations, depletion started at 0:00 UT and ended at 12:00 UT on April 3 (lasting roughly 12 hr), which is consistent with the average dropout period of 16 ± 10 hr (M. J. Weberg et al. 2012). While this is a single-event study, the dropout discovered in the SWA-HIS observations at 0.3 au aligned precisely with the sector boundary, revealing a clear association of the Fe/O dropout with the HCS.

4.4. The Oxygen Preferential Heating and Ionic Charge-state Ratio

T_O/T_p frequently exceeds the super-mass-proportional threshold of 16 throughout the sector associated with active regions. This thermal state displays a positive correlation with the O^{7+}/O^{6+} ratio, suggesting that high-temperature

conditions and preferential oxygen heating mechanisms coexist in the lower corona.

Previous remote-sensing observations revealed that Oxygen (O^{5+}) experiences nonthermal heating over protons in the low solar corona. UVCS observations suggest that Oxygen flows at twice the proton speed with a super-mass-proportional temperature ratio ($T_{O^{5+}}/T_p > 16$) and extreme temperature anisotropies ($T_{\perp}/T_{\parallel} > 10$) by $2 R_s$ (S. R. Cranmer et al. 2008). The oxygen temperatures reaches 3×10^7 K with a ratio of $T_{O^{5+}}/T_p = 21$ by $5 R_s$ in streamers. Such heating could be attributed to preferential energy transfer from Alfvén cyclotron waves as they dissipate to ions in the perpendicular direction (Y. Q. Hu & S. R. Habbal 1999; J. V. Hollweg & P. A. Isenberg 2002; J. C. Kasper et al. 2008). Considering the generally high Alfvénicity in the region of interest (Figure 2(g)), this mechanism could explain the O^{6+} temperature profile in our observations. The presence of ion cyclotron heating at rates that correlate with the turbulent energy cascade rate was found in recent near-Sun observations (T. A. Bowen et al. 2024), implying that the role of cyclotron heating is important for the oxygen heating in the corona, along with other emerging theories compatible with the helicity barrier (J. Squire et al. 2022). Kinetic Alfvén waves are also regarded as viable drivers for the observed ion acceleration and heating at 1–10 R_s (S. Ayaz et al. 2024).

SWA-HIS observation also indicates that the estimated coronal temperature is expected to be high since O^{7+}/O^{6+} remained above typical solar wind values during the HCS crossing. These results are consistent with previous observations at 1 au in the slow-wind regime, which can emerge from several distinct sources, including the high-temperature plasma of helmet streamer stalks (L. Zhao et al. 2009; L. Zhao & L. Fisk 2011), active regions and their peripheries (P. C. Liewer et al. 2003; J. L. Culhane et al. 2014), and complex coronal hole boundaries where magnetic reconnection occurs (S. K. Antiochos et al. 2011; N. U. Crooker & R. L. McPherron 2012). These sources are mostly related to small- or large-scale reconnection, suggesting connections to the HCS.

4.5. The Fully Stripped Carbon Dropouts

SWA-HIS observations revealed depletion of fully stripped carbon (C^{6+}) that correlate with the preferential heating of O^{6+} throughout the sector associated with active regions. This depletion process appears to operate with greater efficiency under conditions characterized by elevated O^{7+}/O^{6+} source ratios.

As a mechanism for fully stripped ion dropouts, J. C. Raymond et al. (2022) proposed that inside relatively dense structures, plasma heating is reduced by localized changes in Alfvén speed, wave reflection, and the associated turbulent cascade for the outlier wind. This process relies on the fact that these fully stripped species share an identical cyclotron frequency with He^{2+} . Within these dense structures, the wave power at the cyclotron frequency of the fully stripped ions is preferentially absorbed by He^{2+} . Consequently, the remaining fully stripped species may not receive sufficient heating to overcome solar gravity. This mechanism consistently explains the features observed in the outlier solar wind, since all of the depleted ion species share the same mass-to-charge ratio, and therefore the same cyclotron frequency (J. C. Raymond et al. 2022). Y. J. Rivera et al. (2021) found that the He/H ratios in

the outlier winds increased compared to regions adjacent to the outliers, and associations with magnetic field variations are suggested. The modeling calculation shows that the depletion process should happen around $1.6 R_{\odot}$ or above (L. Zhao et al. 2017b). Although the depletion of full stripping observed here is not as severe as in these outlier winds, cyclotron wave absorption by He^{2+} appears to be equally applicable to explaining our observations. Furthermore, our results suggest that full carbon stripping is a more ubiquitous phenomenon; as Figure 3(f) demonstrates, the fully stripped carbon dropouts occur even within the normal solar wind regime defined in the previous study by L. Zhao et al. (2017b).

Another key characteristic of these outlier streams is that their source region is identical to that of the normal slow solar wind, with 65% of events linked to active regions. Intriguingly, the observed negative correlation between C^{6+}/C^{5+} and T_O/T_p was maintained across a wide region on one side ($B_r > 0$) of the sector boundary but remains completely absent from the HCS itself. Ion composition observations from SWA-HIS, conducted at unprecedented proximity to the Sun, links the phenomena in Region 4 to the sector with the active regions (Figure 1(b)) bounded by the HCS. Therefore, these features could be direct signatures of heavy ions at the release from the Sun’s surface through interchange reconnection (e.g., L. A. Fisk 2005). However, a single-event study cannot uniquely confirm this pathway and such an interpretation is a plausible scenario to connect the observed correlations and previous outlier wind studies.

5. Conclusion

SWA-HIS observations at 0.3 au have successfully captured the fine-scale elemental and ionic signatures of the inner heliospheric solar wind surrounding the HCS. These single-event time-series observations map directly back to the localized heating and elemental fractionation processes operating at the Sun or in the corona. Our analysis yields three primary observational findings, covering both transient small-scale features and broader sector-level characteristics:

1. Around the TSB, the magnetic topology inferred from the electron strahl changes markedly, yet the heavy-ion composition profiles remain nearly invariant within uncertainties. In contrast, within the observed SDOs upstream of the TSB, where modest increases in proton speed are present, the carbon and oxygen charge-state ratios remain essentially unchanged while T_O/T_p exhibits distinct local drops. Together, these measurements show that small-scale HCS-related structures at 0.3 au can be consistently matched with their heavy-ion thermal and compositional signatures considering differences in freeze-in height and preferential-heating zone along the field line.
2. The reduction of Fe/O, which reaches its minimum in exact alignment with the TSB as the plasma maps toward the coronal loop center. Gravitational settling under extended confinement within closed magnetic structures is suggested to explain this trend. The minimum Fe/O ratio was observed to be $(4.54 \pm 0.54) \times 10^{-2}$ for this crossing event.
3. In the post-HCS active-region sector, T_O/T_p frequently exceeds the super-mass-proportional threshold of 16.

Over this interval, T_O/T_p is positively correlated with the O^{7+}/O^{6+} ratio and negatively correlated with the C^{6+}/C^{5+} ratio, whereas at lower T_O/T_p (primarily in Regions 1–3) the correlation between T_O/T_p and C^{6+}/C^{5+} is weakly positive. These empirical relationships indicate that T_O/T_p can be used to distinguish different regimes of oxygen and carbon charge-state behavior within the same sector.

These unprecedented ionic-composition observations, captured by Solar Orbiter at 0.3 au, provide a detailed, observationally based characterization of fine-scale structure and sector-scale signatures in and around the HCS that was previously accessible only at 1 au. Analyzing charge-state ratios and mass-dependent heating has offered a new way to relate small-scale structures to their thermal and compositional properties. The significant depletion of the Fe/O ratio and its precise alignment with the HCS supply quantitative constraints relevant to models of gravitational settling and heavy-ion fractionation in long-confined large coronal loops. Moreover, because oxygen ions have not yet undergone significant Coulomb thermalization (P. J. Tracy et al. 2015) at 0.3 au, the measured oxygen temperatures can be associated with small-scale signatures originating closer to the Sun than is possible at 1 au. We emphasize that this study is based on a single and rare HCS crossing at 0.3 au. The observational signatures themselves are robust for this event. However, our physical interpretations are necessarily nonunique and should be regarded as hypotheses consistent with these constraints. Testing their generality would require larger statistical samples and dedicated modeling.

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Author Contributions

K.O. came up with the initial research concept, and provided the formal analysis and validation. He also edited the manuscript. R.M.D., F.A., S.T.L., J.M.R., and S.A.S. provided and validated SWA-HIS data, supported data reduction, and edited the manuscript. B.L.A. contributed to the data interpretation and edited the manuscript. A.G. and L.K. supported SWA-HIS data preparation and edited the manuscript. T.S.H. provided and validated MAG data, and edited the manuscript. S.A.L. provided and validated SWA-HIS data, and edited the manuscript. P.L. provided and validated SWA-

PAS data and edited the manuscript. C.J.O. led SWA team, provided and validated SWA-EAS data, and edited the manuscript.

Data availability

All the data analyzed in the manuscript are publicly available online. URL links are provided in the methodology section. Correspondence and requests for materials should be sent to Keiichi Ogasawara.

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